

ASSIGNMENT BRIEF

HTU Course No: 10204210	HTU Course Name: Data Analytics
BTEC Unit Code: F/618/7415	BTEC UNIT Name: Data Analytics



Student Name/ID Number/Section	
HTU Course Number and Title	10204210 Data Analytics
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Academic Year	2025-2026 2
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Assignment Title	Emergency Response Analytics and Decision Optimization
Assignment Ref No	1
Issue Date	19/04/2026
Formative Assessment dates	From 19/04/2026 to 04/06/2026
Submission Date	13/06/2026
IV Name & Date	Batool Alarmouti 18/04/2026

Submission Format

Practical Assignment Submission:

Submit to the university's eLearning system via <https://elearning.htu.edu.jo> by the specified deadline. The submission must include the following:

1. A Python source code file (.ipynb) for descriptive analytics.
 2. A Python source code file (.ipynb) for predictive analytics.
 3. Python source code files (.ipynb), and figures detailing the experiments and results specific to the perspective analytics.
- No compressed files or folders (no .zip or .tar extensions).
 - Plagiarism will result in course failure.

Report Submission:

- Submit to the university's eLearning system via <https://elearning.htu.edu.jo> by the specified deadline.
- The submission must be a PowerPoint file (.ppt/pptx) only.
- No compressed files or folders (no .zip or .tar extensions).
- Plagiarism will result in course failure.
- A signed declaration form is required

Oral Exam:

- An oral discussion will be scheduled with your instructors to assess your understanding of the assignment.
- Answering all questions in the report will be evaluated through the oral assessment. So, if any question in the oral assessment for any criteria is not answered at the required level, that means you will lose that criterion.

Unit Learning Outcomes

LO1 Discuss the theoretical foundation of data analytics that determine decision-making processes in management or business environments

LO2 Apply a range of descriptive analytic techniques to convert data into actionable insight using a range of statistical techniques

LO3 Investigate a range of predictive analytic techniques to discover new knowledge for forecasting future events

LO4 Demonstrate prescriptive analytic methods for finding the best course of action for a situation

Assignment Brief and Guidance

You are employed as a Data Analyst at an advanced emergency response coordination centre. The organization is facing challenges related to delays in response times, inefficient dispatch decisions, and complex decision-making under dynamic conditions (such as traffic, weather, and incident severity). To address these challenges, you are provided with the Integrated Emergency Response Analytics Dataset. Your task is to develop a comprehensive analytical solution that improves operational understanding, forecasting, and decision-making. In this project, you should demonstrate your ability to use Python in implementing the analysis. Throughout your work, you are expected to demonstrate a clear understanding of the data analytics process and its role in supporting decision-making.

You need to present the impact of data analytics on improving the decision-making process by **determining the activities, techniques, and tools** used in data analytics, and exploring the **roles of descriptive, predictive, and prescriptive analytics within this emergency response scenario**. In addition, **evaluate the importance of data analytics in improving operational decision-making for the emergency centre**.

As a data analyst, your primary responsibility is to explore the dataset to gain an understanding of historical emergency response operations. You are required to complete the following activities to apply three data analytics types:

Activity 1: Understanding the Operational Context through Data

You are required to identify various descriptive analytics techniques that can be applied using Python to identify patterns, trends, and relationships among variables such as response time, traffic conditions, weather, incident severity, and resource availability. You should present the insights that explain how these factors influence emergency response performance and communicate your findings clearly using appropriate measures, visualizations, and interpretations.

Activity 2: Predicting Emergency Response Time

To support proactive planning, the emergency response centre requires the ability to predict future response performance. You are required to identify the predictive techniques that can be used to help to predict the **response time** for future incidents. You must:

- Apply at least two feature selection techniques to identify the most significant predictors.
- Implement at least three predictive models using Python.
- Compare the models' performances using appropriate regression metrics.

You need to evaluate the performance of the used models to identify the most effective model. And how predictive analytics can support forecasting and improve planning in emergency response operations.

Activity 3: Optimizing Emergency Response Decisions

For each emergency incident, the response centre must determine the optimal dispatch strategy that *minimizes response time* by leveraging the most predictive numerical features from historical incident data. You need to apply an optimization algorithm to learn the best weighting coefficients that correlate key operational factors with response time, enabling data-driven dispatch decisions.

Decision Variables

Define the following decision variables:

- **w1:** Weight coefficient for Distance to Incident
- **w2:** Weight coefficient for Number of Injuries
- **w3:** Weight coefficient for Hospital Capacity
- **w4:** Weight coefficient for Fuel Level
- **b:** baseline Intercept (minimum response time)

Constraints

The decision-making process must satisfy the following constraints:

- 1. Distance Coefficient Constraint:** The distance travelled has a direct positive relationship with response time; each kilometre travelled should add between 0.2 and 0.8 minutes to response time.
- 2. Injury Coefficient Constraint:** More injured persons require additional resources, which increases response time; each additional injury adds 0.1 to 0.5 minutes to the response time.
- 3. Hospital Capacity Constraint:** Higher hospital capacity enables faster patient acceptance, reducing effective response time; when hospital capacity is low (approaching 0%), this penalty adds up to 3 minutes.
- 4. Fuel Level Constraint:** Lower fuel levels may require refuelling or cause delays; when the fuel level is critically low (approaching 0%), this penalty adds up to 4 minutes.
- 5. Baseline Constraint:** Every emergency response has a minimum unavoidable dispatch time; Base dispatch and preparation time ranges from 5 to 10 minutes.
- 6. Non-Negativity Constraint:** All weight coefficients must be non-negative

You are required to use the following formulation to predict the response time for each incident. The fitness function is defined as the negative value of the prediction error across all incidents, computed using a regression error metric.

Response Time = (w1 x Distance_to_Incident) + (w2 x Number_of_Injuries) + (w3 x Hospital_Capacity) + (w4 x Fuel_Level) + b

You are required to analyze the result of the Genetic algorithm by:

- Determine the best solution for each decision variable.
- Show the final Response Time formula with the optimal weight coefficients.
- Evaluate the performance of your solution by determining the average Predicted Response Time, average Actual Response Time, and the difference between actual and predicted response time for all the incidents.
- Describe and justify how the optimized solution helps the emergency response centre determine the best course of action under different conditions.
- Demonstrate how the solution using GA improves decision-making and operational efficiency by evaluating the algorithm's convergence in terms of fitness score and average response time.

Learning Outcomes and Assessment Criteria			
Learning Outcome	Pass	Merit	Distinction
LO1 Discuss the theoretical foundation of data analytics that determine decision-making processes in management or business environments	P1 Identify data analytic activities, techniques, and tools. P2 Demonstrate an ability to use a popular programming language or tool used in the data analytics industry.	M1 Investigate the three types of data analytic methods and their use in industry. M2 Show how these descriptive analytic techniques contribute to decision-making.	D1 Evaluate the importance of data analytical techniques to the decision-making process.
	P3 Investigate descriptive analytic techniques and explain with appropriate examples. P4 Apply an appropriate tool or programming language to demonstrate these descriptive analytics techniques.		
LO3 Investigate a range of predictive analytic techniques to discover new knowledge for forecasting future events	P5 Identify predictive analytic techniques and describe these techniques with examples. P6 Apply an appropriate tool or programming language to demonstrate these predictive analytic techniques.	M3 Compare a range of predictive analytical techniques for forecasting purposes.	D2 Evaluate how predictive analytic techniques can be used for forecasting purposes.
LO4 Demonstrate prescriptive analytic methods for finding the best course of action for a situation	P7 Analyse prescriptive analytic techniques with appropriate examples. P8 Demonstrate these techniques using an appropriate programming language or tool.	M4 Describe how these prescriptive analytic techniques are used to find the best course of action in a situation.	D3 Apply an appropriate programming language or tool to demonstrate how these prescriptive analytic techniques are used to find the best course of action in a situation.